
SMART VISIT

From 5,100 places to one coherent itinerary

Personalized tourist itinerary recommendation
based on user preferences and contextual data

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Master D2SM · ENSAM Meknes · 2025-2026

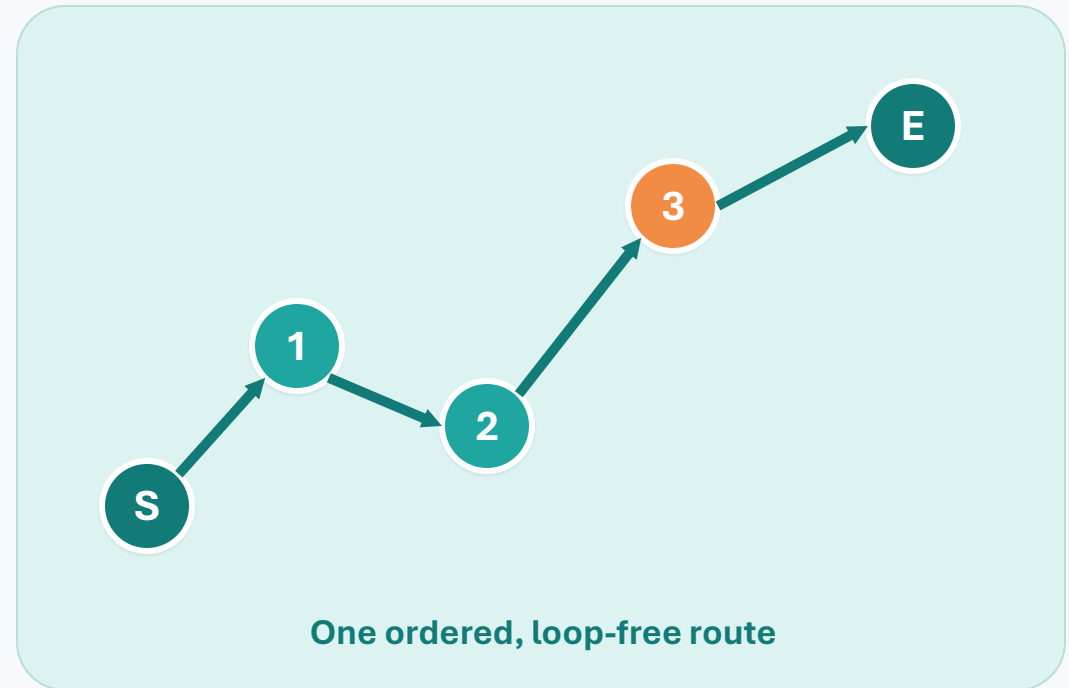


A good trip answers two questions: what to visit, and in what order?

1 WHAT TO VISIT



2 IN WHAT ORDER



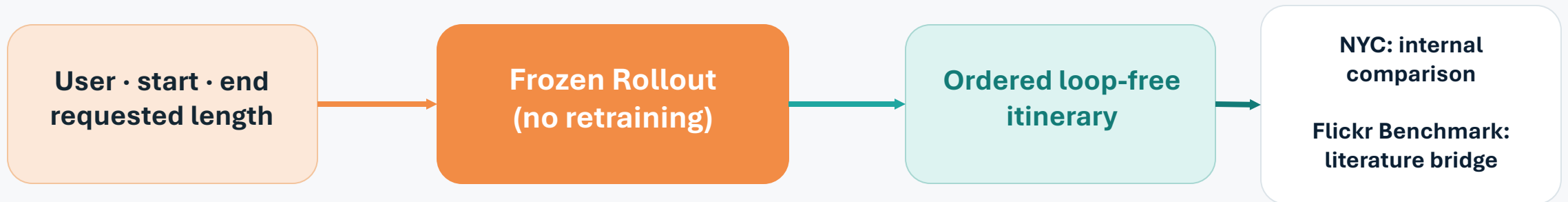
Conceptual illustration based on the thesis problem definition.

One engine, one decoder, and two fair evaluations

PHASE 1 • NEXT-POI



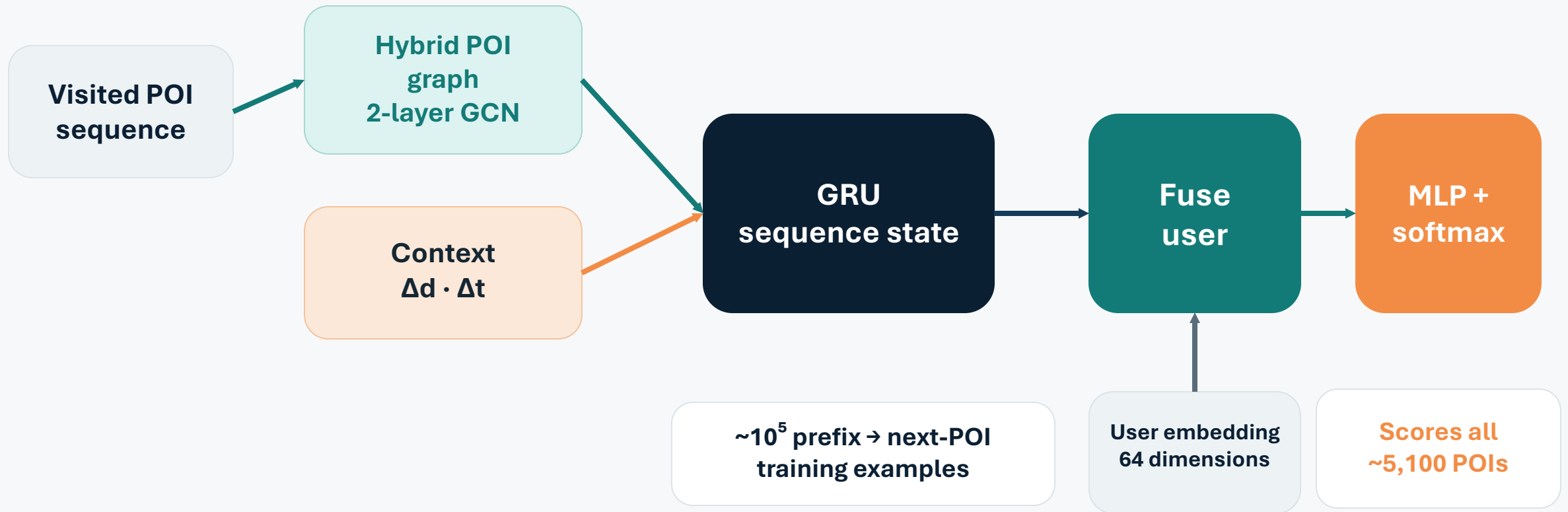
PHASE 2 • ITINERARY



Never mix scores across tasks, datasets, protocols, or metrics

Source: Thesis Sections 3.2 and 3.6. Two tasks are evaluated separately.

The engine combines places, context, and user preferences



Source: Thesis Sections 3.3-3.4. Implemented context is distance and elapsed time.

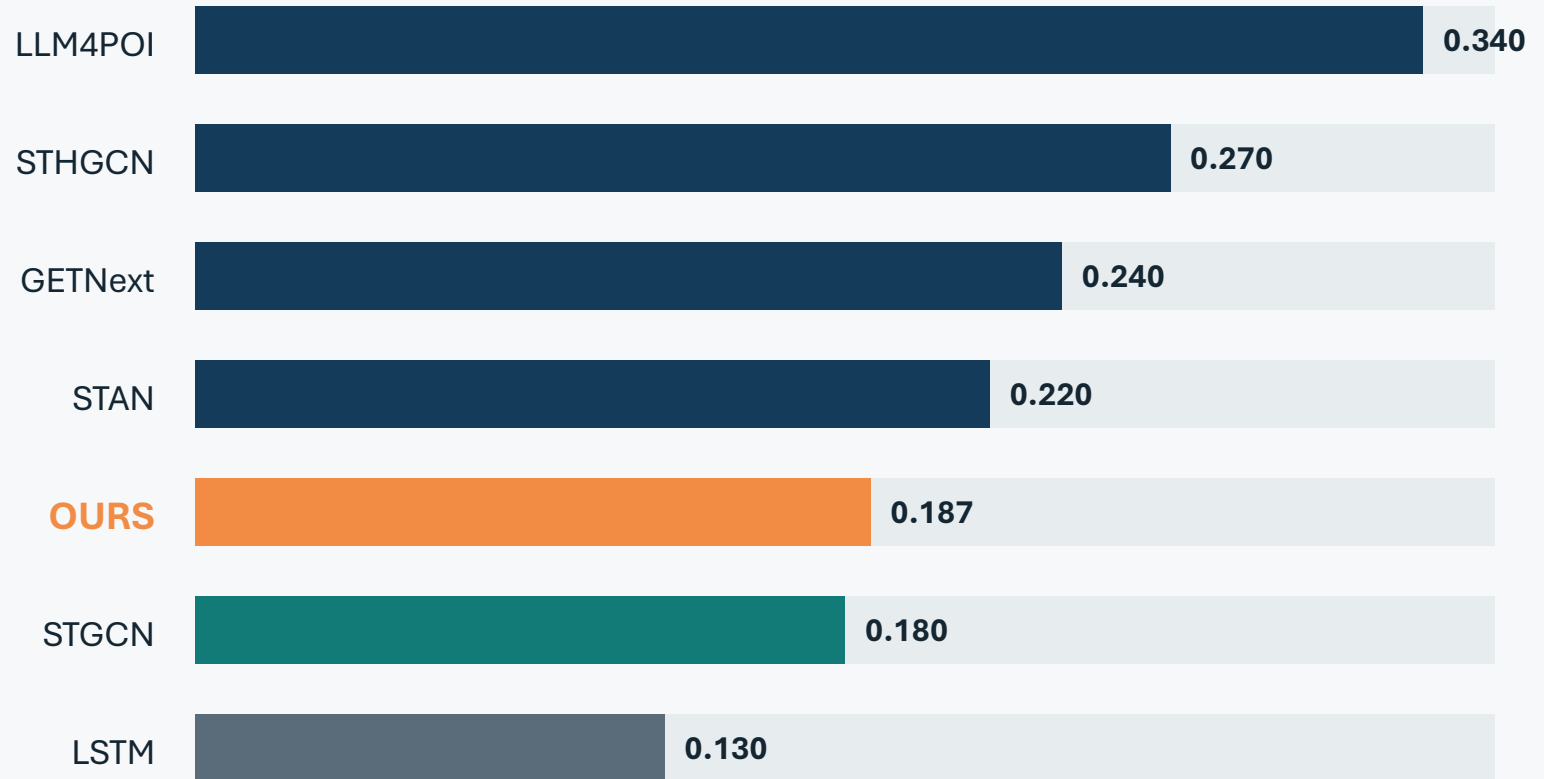
The engine is competitive, but not state of the art

0.187

HR@1

Exact next POI ranked
first
~1 time in 5 among
~5,100

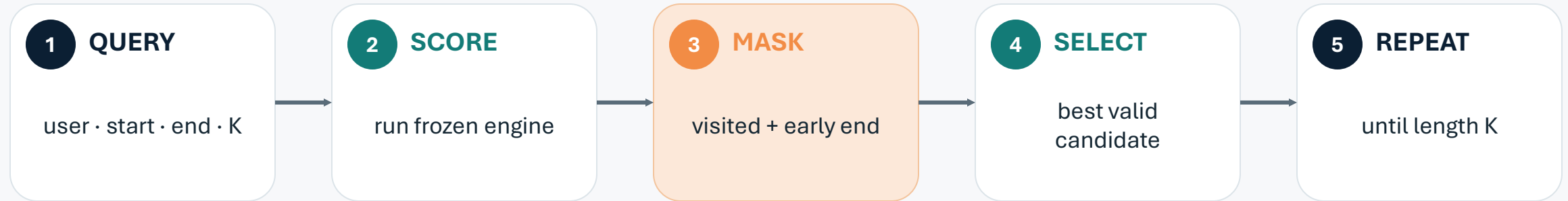
HR@10 = 0.588



Higher is better · full vocabulary · Foursquare NYC

Source: Thesis Table 3.2 and Figure 3.2. Published comparison values use the same NYC full-vocabulary HR@1 protocol.

Frozen Rollout turns next-POI scores into a loop-free route



The route grows one valid stop at a time



Source: Thesis Section 3.5. Decode-time constraints: loop-free, fixed endpoints, requested length.

Dense supervision beats a purpose-trained itinerary model

FROZEN ROLLOUT

$\sim 10^5$

prefix → next-POI examples

Reuses a converged engine

0.289

pairs-F1

>

TRAINED POINTER

1

whole trajectory per session

Trains end-to-end from scratch

0.259

pairs-F1

NYC · n = 2,880 · length ≥ 3 · order-aware pairs-F1

Source: Thesis Table 3.3. Beam search changes Frozen Rollout only from 0.289 to 0.290.

A fair literature comparison requires the same benchmark

NYC INTERNAL STUDY

~5,100

POIs

NYC pairs-F1
0.26 - 0.29

≠

FLICKR BENCHMARK

27 - 88

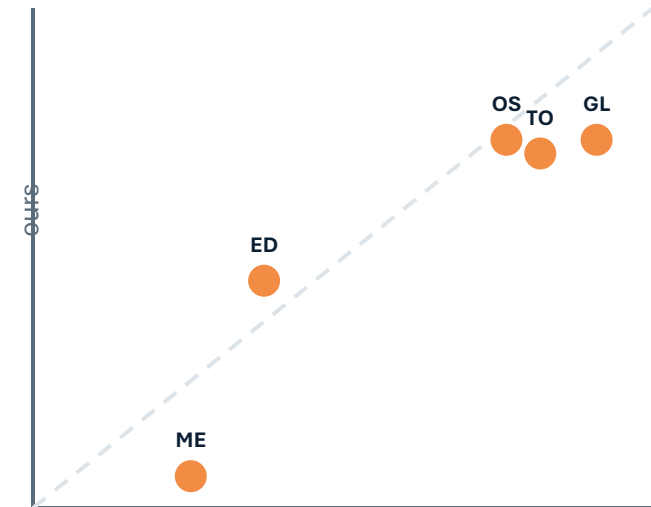
POIs per city

Published pairs-F1
~0.26 - 0.85

THE BRIDGE

Run our methods on their data, protocol, and metric

Random reproduction check

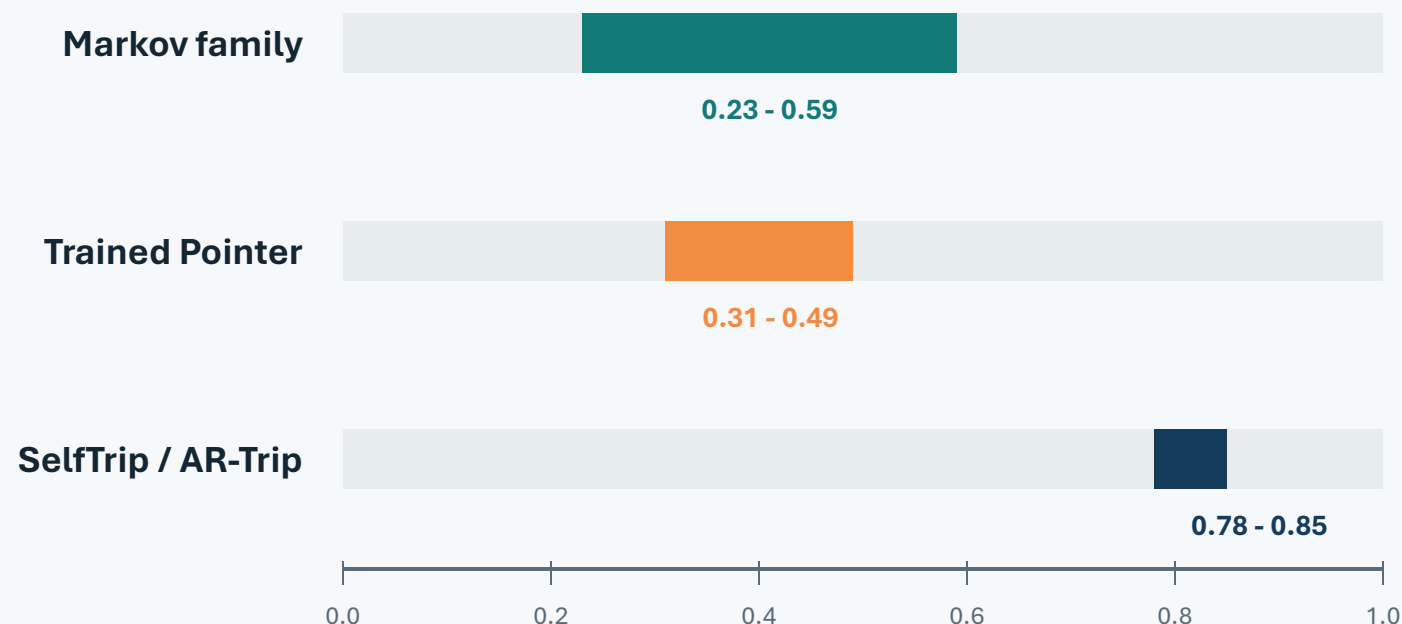


maximum absolute gap = 0.021

Source: Thesis Tables 3.4-3.5 and Figure 3.4; Chen et al. (2016). Random uses the Chen-2016 protocol.

The benchmark confirms our position - and exposes cold-start

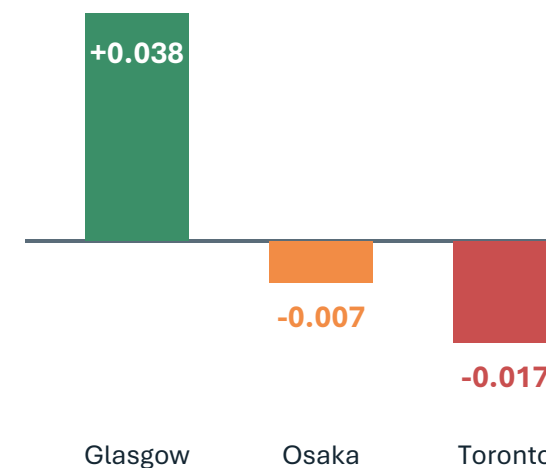
FLICKR BENCHMARK · PAIRS-F1 ON THE PUBLISHED SCALE



Our classical methods reach the published scale; neural SOTA remains clearly ahead

USER EMBEDDING ABLATION

Personalization helps only when enough user history survives leave-one-out



Separate 30-epoch ablation

Source: Thesis Tables 3.6-3.8 and Figure 3.5. The ablation is separate from the main Trained Pointer run.

THREE RESULTS TO REMEMBER

A defensible path from prediction to itinerary

0.187 HR@1

0.289 > 0.259

max gap 0.021



honest next-POI engine
~5,100 NYC candidates

Frozen Rollout beats
the Trained Pointer

Flickr Benchmark reproduces
Chen 2016 Random

Compare only when dataset + protocol + metric are identical

Data preparation and evaluation protocols

PHASE 1 · FOURSQUARE NYC

- 1 **Raw data** ~227K check-ins · ~1.08K users · ~38K venues
- 2 **Filter** Iterative user/POI count ≥ 10
- 3 **Final catalog** ~5,100 POIs · ~1,000 users
- 4 **Sessions** Split after >24 h silence
- 5 **Split** Chronological per user: 70 / 10 / 20
- 6 **Evaluation** Full vocabulary · no sampled negatives

PHASE 2 · FLICKR BENCHMARK

City	POIs	Users	Traj.	Eval. ≥ 3
Toronto	29	1,395	6,057	335
Osaka	27	450	1,115	47
Glasgow	27	601	2,227	112
Edinburgh	28	1,454	5,028	634
Melbourne	88	1,000	5,106	442

Chen-2016 protocol

Leave one trajectory out · first + last POI + length given
length ≥ 3 · loop-free · pairs-F1

Metric definitions and complete result tables

NEXT-POI · NYC

Metric	Score
HR@1	0.187
HR@5	0.476
HR@10	0.588
NDCG@5	0.339
NDCG@10	0.375
MRR	0.316

FLICKR BENCHMARK · PAIRS-F1

Method	Toronto	Osaka	Glasgow	Edin.	Melb.
Random	.298	.301	.301	.270	.227
PoiPopularity	.443	.413	.510	.439	.320
Markov	.504	.421	.587	.449	.333
MarkovPath	.528	.398	.543	.452	.346
Trained Pointer · beam 3	.431	.399	.489	.414	.312

ITINERARY · NYC

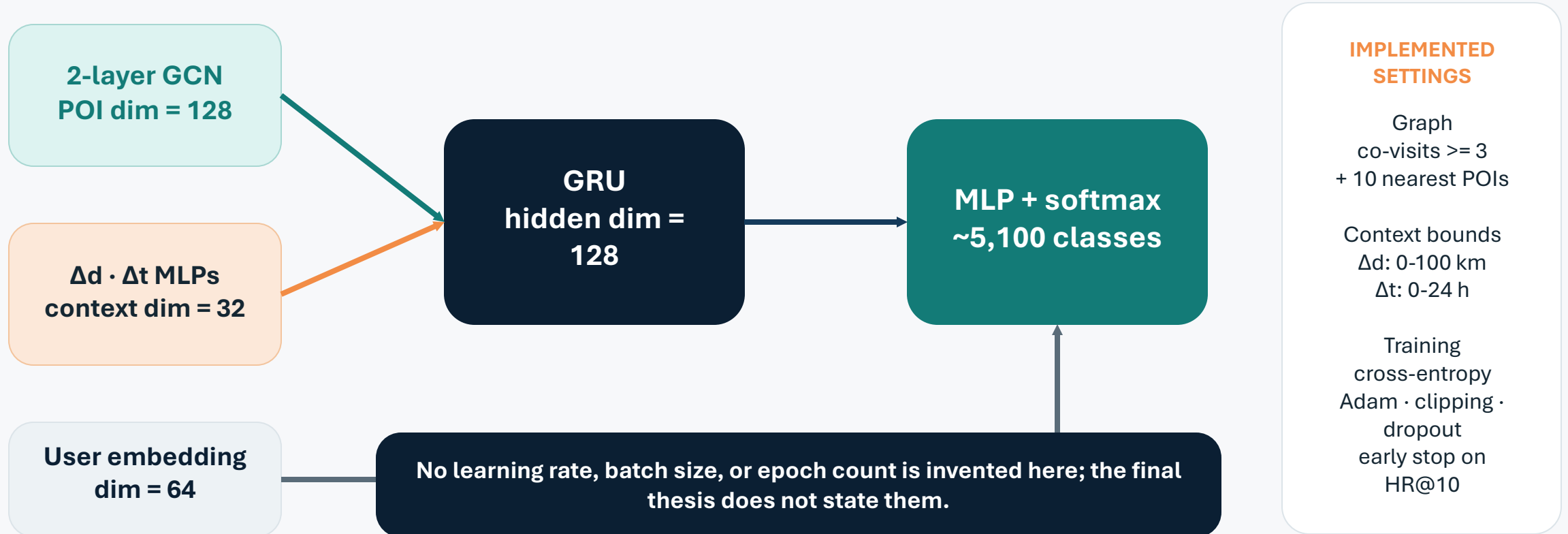
Method	pairs	set	exact
Frozen Rollout · greedy	.289	.609	.054
Frozen Rollout · beam 3	.290	.610	.057
Trained Pointer · v1	.259	.578	.043
Trained Pointer · v2 +ctx	.261	-	-

Source: Thesis Tables 3.2, 3.3 and 3.6. Dashes mean unreported, not zero.

METRIC BOUNDARY

HR@k / NDCG / MRR rank one next POI.
set-F1 measures recovered places. pairs-F1 measures order.
These values never share one leaderboard.

Detailed architecture, dimensions, and training controls



Source: Thesis Sections 3.3-3.4. Co-visitation edges are computed from training data only.

Why the Trained Pointer loses - limitations and next steps

TRAINING CONTROLS

0.259 → 0.261

Adding context recovers only +0.002

- ✓ validation pairs-F1 reaches a real peak
- ✓ early stopping applied
- ✓ context-aware variant tested
- ✓ Trained Pointer also trails Markov on the Flickr Benchmark

Conclusion: a genuine negative result for this tested setup, not a universal claim about all itinerary models.

MAIN LIMITATIONS

- below neural / LLM SOTA
- scorer remains locally myopic
- context only uses Δd and Δt
- decode-time Δt is assumed
- no opening hours or time windows
- one next-POI city
- offline evaluation only
- no deployment or user study

NEXT STEPS

- weather and time-of-day context
- persona/content cold-start encoder
- travel cost and opening hours
- schedule-aware constrained decoding
- pre-training and trajectory augmentation
- multi-city validation
- user study

Orienteering / ILS: scoped, NOT implemented

Source: Thesis Sections 3.7.5 and General Conclusion. Reader-facing method names are used throughout.